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Questions from the Lecture

1. Conceptual Question:

What is the key difference between L1 (LASSO) and L2 (Ridge) regularization, and how does it affect model coefficients?

2. Analytical Question:

How does increasing the regularization strength (λ) affect the bias-variance tradeoff in a machine learning model?

3. Practical Question:

Why is it important to scale features before applying regularization, and how does it impact the interpretation of model coefficients?

Takeaways from the Lecture

1. Understanding Overfitting:

Overfitting occurs when a model performs exceptionally well on training data but poorly on unseen data. Regularization is a critical tool to combat overfitting by constraining model complexity.

2. Role of Regularization Techniques:

- L1 (LASSO) regularization introduces sparsity by driving weak feature coefficients to zero, making it ideal for feature selection.
- L2 (Ridge) regularization uniformly shrinks coefficients, preventing overfitting without eliminating features.

3. Importance of Hyperparameter Tuning:

The regularization strength (λ) must be carefully tuned (e.g., using cross-validation) to balance model bias and variance, ensuring optimal performance on unseen data.